

FreshRoute: An AI-Driven Smart Platform for Reducing Waste and Improving Efficiency in Sri Lanka's Fruit Supply Chain

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Abstract- Sri Lanka's fruit supply chain faces persistent inefficiencies caused by poor harvest planning, intermediary dependency, transportation spoilage, and quality disputes between farmers and buyers. This paper presents FreshRoute, an integrated digital platform designed to improve transparency, efficiency, and trust across the fruit supply ecosystem. The system combines AI-driven analytics, blockchain-based transaction management, optimized logistics planning, and automated quality verification to support data-informed decision making throughout the supply chain. The platform is implemented using a scalable technology stack including machine learning models for demand forecasting, smart contracts for secure trading records, route optimization algorithms for transportation planning, and computer vision techniques for fruit quality assessment and complaint validation. Module-level evaluation demonstrates promising performance across forecasting accuracy, logistics efficiency, and quality detection reliability. The user interface is designed to support farmers, buyers, and logistics providers with intuitive dashboards, alerts, and traceability features suitable for users with limited digital literacy. In addition, FreshRoute incorporates commercialization pathways such as premium analytics services, logistics partnerships, and integration with economic centers to ensure long-term sustainability. The architecture provides a practical foundation for reducing post-harvest losses, improving farmer income stability, and modernizing Sri Lanka's fruit supply chain through a unified digital ecosystem.

Keywords- smart agriculture, fruit supply chain, demand forecasting, blockchain, logistics optimization, computer vision, Sri Lanka

I. INTRODUCTION

Fruit production remains economically significant in Sri Lanka; however, the value chain is characterized by substantial inefficiencies and high levels of post-harvest loss. A national review reports losses of approximately 30–40% for fruits, attributing them to inadequate handling practices, poor storage conditions, limited cold chain integration, and persistent information gaps that expose small scale producers to avoidable waste [1]. Commodity-level studies further highlight the severity of the issue. For instance, a life-cycle assessment of banana supply chains in Sri Lanka identified total post-harvest losses of 27.23% from harvesting to retail, with 7.61% occurring during transportation between wholesale and retail stages [2]. Additionally, official reports indicate that approximately 21% of fruits are wasted annually during transportation due

to inappropriate handling practices [3], while the National Institute of Post Harvest Management (NIPHM) identifies unsuitable packaging as a major contributing factor and has introduced interventions such as subsidized crate programs and farmer training initiatives [4].

These losses do not originate from a single bottleneck but rather from fragmented decision-making across the supply chain. Smallholder farmers frequently determine harvest timing with limited visibility into future demand and price trends. Market transactions remain influenced by intermediaries, such as collectors and commission agents, who maintain informational and bargaining advantages. Following trade execution, transport planning rarely accounts for differences in perishability or handling requirements among fruit consignments. Even at the final stage, manual quality inspection processes introduce uncertainty, enabling false complaints and unresolved disputes. Consequently, a single fruit consignment typically passes through four disconnected decision spaces planning, trading, logistics, and quality assurance with minimal data integration or verifiable accountability.

Existing research addresses these decision spaces individually rather than as an integrated system. Time-series forecasting models, including SARIMA, Prophet, and long short-term memory (LSTM) networks, support short-term demand prediction [5]–[7]. Permissioned blockchain platforms and smart contracts provide mechanisms for transparent and auditable transactions [8], [9]. Studies on perishable vehicle routing demonstrate that freshness-aware logistics can outperform traditional distance-based optimization approaches [11]. Similarly, advancements in computer vision have significantly improved fruit grading and defect detection capabilities [12], [13]. However, the isolated application of these technologies fails to resolve the broader issue of operational fragmentation. Forecasting models that do not influence trading decisions, marketplaces that are disconnected from logistics planning, and quality assessment systems that are not integrated with complaint workflows deliver only partial benefits.

This paper introduces FreshRoute, an integrated digital platform designed to modernize the fruit supply chain in Sri Lanka. The proposed system unifies four core modules: demand forecasting, a permissioned blockchain-based marketplace, perishability-aware logistics optimization, and AI-driven quality verification. The primary contributions of

this work are threefold. First, it integrates multiple research domains into single, technically coherent architecture tailored for smallholder environments. Second, it defines a structured data flow across modules, enabling upstream decisions to be validated through downstream outcomes. Third, it establishes an evaluation framework designed for phased pilot deployment, emphasizing real-world applicability rather than purely laboratory-based performance assessment.

II. BACKGROUND

A. Forecasting and farmer-facing intelligence

Forecasting studies consistently demonstrate the effectiveness of modern time series models, particularly in scenarios where seasonality and non-linear interactions significantly influence demand patterns. The Prophet model proposed by Sean J. Taylor and Benjamin Letham remains widely adopted due to its ability to provide interpretable trend and seasonal decomposition [5]. Similarly, research by Tae Woo Yoo and In-Sik Oh demonstrates that seasonal long short-term memory (LSTM) models outperform traditional such as Auto ARIMA, Prophet, and standard LSTM variants in agricultural sales forecasting tasks [6]. Comparative analyses between ARIMA and LSTM models further indicate that recurrent neural networks are generally more effective in capturing complex temporal dependencies [7]. Despite these advancements, much of the existing literature focuses primarily on predictive accuracy as the final objective. Limited attention has been given to the practical deployment of forecasting outputs within real world agricultural contexts. In particular, the “last mile” challenge translating model predictions into actionable harvest decisions remains insufficiently addressed. This gap is especially significant for smallholder farmers operating under constraints such as intermittent connectivity, limited datability and delivery of predictive insights are as critical as the accuracy of the models themselves.

B. Blockchain and direct exchange

In agri-food supply chains, blockchain is most frequently proposed as a traceability mechanism. Kamilaris et al. review a broad set of initiatives and conclude that blockchain can improve transparency, but widespread use remains constrained by technical, institutional, and adoption barriers [8]. Hyperledger Fabric is especially relevant in enterprise contexts because it supports permissioned membership, modular consensus, and private data exchange between known participants [9]. More recent work extends this discussion by showing how decentralized identity can support verifiable product identity in agri-food systems [10]. Even so, the dominant use case remains origin tracing or certification. Less attention has been given to using permissioned ledgers as the operational core of a direct farmer-to-buyer marketplace with payment logic, reputation updates, and linkage to logistics and quality events.

C. Perishable logistics under local constraints

Research on fresh-produce logistics highlights a persistent gap between theoretical optimization models and real-world operating conditions. Utama et al. demonstrate that vehicle-routing problems for perishable goods are typically formulated using constraints related to freshness, timing, and cost [11]. While these models are effective in controlled environments, they often assume a level of availability and control over fleets, cold-chain infrastructure, and telemetry that is not present in fragmented smallholder

systems. Empirical evidence from Sri Lanka reflects a contrasting reality, where significant losses continue to occur due to transport delays, heat exposure, inadequate packaging, and limited access to suitable vehicles [1]–[4]. These challenges are further compounded by limited coordination among stakeholders and the absence of real-time decision-support mechanisms in rural contexts. Consequently, practical system design must adopt lightweight, adaptive routing strategies that explicitly account for infrastructural limitations rather than treating them as fixed assumptions.

D. Quality verification beyond classification

Computer vision has advanced significantly to support automated visual inspection of fruit quality. Reviews of convolutional neural network (CNN)-based approaches report strong performance in tasks such as grading, detection, and classification [12]. More recent deep learning models further enhance defect detection accuracy and reliability in quality assessment [13]. However, supply-chain quality verification extends beyond classification alone. It requires reliable capture of pre-delivery and post-delivery evidence, safeguards against duplicate or manipulated images, and an integrated complaint workflow that links model outputs to user accountability. Techniques such as perceptual hashing offer an effective solution for detecting duplicate or near-duplicate images [14]; however, these mechanisms are rarely incorporated within a comprehensive marketplace framework.

E. Gap synthesis

Literature consistently highlights a critical gap: the absence of unified, smallholder-oriented architecture that integrates forecast-driven harvesting, transparent and auditable market exchange, perishability aware logistics execution, and evidence-based complaint management. Existing solutions largely address these components in isolation, limiting their practical effectiveness. FreshRoute addresses this gap by conceptualizing the fruit supply chain as a closed-loop decision system, enabling seamless data flow and coordination across all stages rather than treating them as independent software functionalities.

TABLE I. CAPABILITY COMPARISON: FRESHROUTE vs EXISTING PLATFORMS

Feature	Govijana	Helaviru	CIC Agri	AgroLink	NIPHM	FreshRoute
AI Demand Forecasting	No	No	No	No	No	YES
Direct Farmer-Buyer Trade	No	Yes	No	Yes	No	YES
Blockchain Traceability	No	No	No	No	No	YES
Perishability Routing	No	No	Part.	No	No	YES
AI Quality Verification	No	No	No	No	No	YES
IoT Real-Time Monitoring	No	No	Part.	No	No	YES
Complaint Validation	No	No	No	No	No	YES
SMS Support	No	No	No	No	No	YES
Smallholder Accessibility	Part.	Part.	No	Part.	Part.	YES

III. METHODOLOGY

Fig. 1 illustrates the integrated FreshRoute architecture. The design separates domain-specific services but uses shared identifiers for producing lots, orders, delivery events, and users. This allows the system to carry a fruit batch from pre-harvest planning to post-delivery dispute resolution without losing contextual data, ensuring traceability, consistency, and seamless data flow across all interconnected modules within the platform.

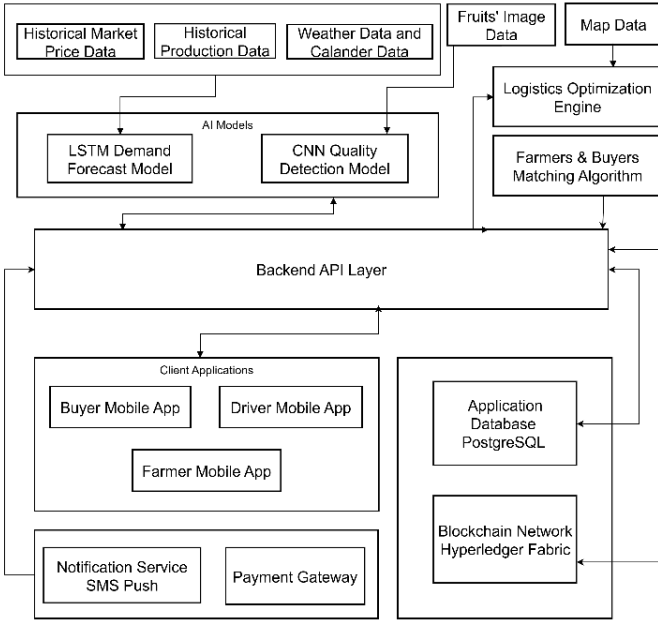


Fig. 1. FreshRoute integrated architecture and closed-loop data flow.

A. Demand forecasting module

The forecasting module ingests historical market prices, historical production data, and weather forecasts. Data preparation includes missing-value treatment, normalization, lag construction, and calendar features. The primary predictive model is a multivariate LSTM that learns short-horizon demand and price movements from temporally ordered observations, while Prophet and SARIMA-style baselines are retained for comparison and fallback behavior [5]-[7].

FreshRoute does not treat forecasting as an analytical endpoint. Model outputs are translated into farmer-facing advisories such as expected demand level, anticipated price direction, and recommended harvest timing windows. These advisories are delivered through a mobile dashboard with local caching and through SMS or push notifications depending on device capability. In this way, predictive intelligence is operationalized as a planning decision rather than a static visualization.

To enhance forecasting accuracy and robustness, the module integrates multiple external and internal data streams. These include market transaction records from regional economic centers, international agricultural indicators from World Bank datasets, and supplementary information obtained through external APIs such as weather services and agricultural statistics platforms. By combining these heterogeneous data sources, the system captures seasonal patterns, supply variations, and market volatility that influence fruit demand.

B. Automated 5-Factor Matching and Fractional Allocation

To disintermediate the supply chain, FreshRoute employs a deterministic matching algorithm that replaces manual brokering. It first enforces an asymmetric quality filter: sub-standard grades are rejected, exact matches score 1.0, and premium grades fulfilling lower-tier orders score 0.7 to maintain market liquidity. Eligible candidates are then ranked using a multivariate function:

$$S_{\text{final}} = 0.35D + 0.20Q + 0.15R + 0.15T + 0.15V \quad (1)$$

This function optimally weighs Haversine spatial distance (D), quality match (Q), decentralized reputation (R), harvest date proximity (T), and volume fulfillment ratio

(V). To prevent inventory lockouts, a "Smart Stock Split" concurrency model dynamically fractionalizes matched yields. It secures the exact required volume required for the active order while instantly releasing the remaining balance back to the public pool, ensuring small volume buyers do not monopolize large-scale farmers.

C. Trustless Escrow and Stateful Fail Safes

Financial security is enforced through a dynamic trustless escrow integrating tokenized pre-approvals (e.g., PayHere). Upon match acceptance, buyers commit a 25% deposit calculated against the current day's FreshRoute price. To account for agricultural price volatility, the final settlement is dynamically adjusted to reflect the platform's calculated FreshRoute price on the agreed pickup date, rather than the initial commitment date. Crucially, this remaining balance is automatically and instantly released to the farmer via smart contract the moment the AI module verifies the fruit quality at the point of farm pickup. To protect producers from uncommitted buyers, the algorithm utilizes a strict 48-hour stateful timeout. If payment tokenization fails within this window, the smart contract invalidates the proposal and releases the reserved stock back to the open market. This programmatically eliminates counterparty risk, ensures immediate liquidity for farmers, and removes the need for human intermediaries.

D. Permissioned marketplace and reputation

The marketplace operates on a permissioned trading layer using Hyperledger Fabric, as agricultural supply chains require controlled onboarding and selective data visibility. To optimize performance, the system employs dual-storage architecture: general queries route to a fast-read Application DB, while high-value state transitions (escrow commitments, delivery confirmations) are immutably locked via smart contracts. Trust is formalized through a Certificate Authority (CA) and decentralized identity principles. Rather than storing private keys centrally, the system issues verifiable cryptographic certificates to users, guaranteeing non-repudiation transaction. Furthermore, successful deliveries, validated complaints, and temperature breaches are appended as signed records to a user's digital profile. This effectively substitutes informal social trust and opaque intermediary judgment with a portable, mathematically verifiable performance history.

E. Perishability-aware logistics optimization

Once an order is matched, the logistics module plans collection and delivery using perishability-aware rules. For each lot, FreshRoute evaluates biochemical and environmental constraints, comparing live weather data and travel distance against fruit-specific temperature thresholds and ethylene gas compatibility. Vehicle assignment then prioritizes limited cold-chain assets for the most vulnerable consignments, strictly separating gas-emitting and sensitive fruits, while non-refrigerated or covered vehicles are reserved for less sensitive loads or shorter trips.

Route generation combines greedy nearest-neighbor logic with a 2-Opt heuristic refinement. The effective transport sequence is optimized using a pre-computed spatial distance matrix, strictly enforcing pickup-before-drop precedence and driver shift limits rather than distance alone. This follows the insight from perishable routing research that freshness preservation and distribution cost must be jointly optimized [11]. Because full-scale IoT deployment is expensive, the module uses lightweight telemetry such as GPS position, timestamps, and simple temperature alerts to trigger instant

escalation when a shipment deviates from acceptable conditions.

Capacity and biochemical compatibility rules are treated as first-class constraints rather than advisory notes. Loads that require strict thermostat locking or ethylene separation from mixed consignments are evaluated before dispatch, and the algorithm marks a vehicle assignment as invalid if these prerequisites or physical weight limits are breached. This is important in Sri Lanka, where unsuitable handling and overstacking remain major drivers of damage [3], [4]. Embedding strict fractional loading and compatibility checks into route creation ensures that fulfillment quality is not delegated entirely to manual discipline.

F. AI-based quality verification and complaint validation

The quality module creates a tamper-resistant record of product condition at dispatch and receipt. Farmers capture images during packing using an in-app camera, and buyers capture images at delivery through the same constrained workflow. Gallery uploads are disabled to reduce the reuse of stale or unrelated images. Alongside the image, the system records metadata including time, location, device ID, order ID, and user identity.

A transfer-learning pipeline based on modern CNN backbones classifies fruit condition and compares pre-dispatch and post-delivery evidence [12], [13]. Perceptual hashing is used to flag duplicate or near-duplicate uploads that may indicate fraud or careless reuse [14]. When a buyer files a complaint, they can upload image evidence and submit a report describing the issue. Administrators then review the complaint and verify the evidence using the same classification model employed during pickup quality verification. By comparing the complaint images with the recorded pre-dispatch images and associated metadata, administrators can determine whether the reported quality issue is valid. This approach keeps the decision process transparent while still benefiting from automated image analysis to assist human review.

Model development should prioritize fruit categories that dominate early pilots and should include images captured under real packinghouse and roadside delivery conditions rather than ideal laboratory backgrounds. Transfer learning is appropriate because labeled local datasets are initially limited. As the system matures, manually reviewed cases can be recycled into the training set so that model behavior improves alongside deployment.

G. Closed-loop integration

The integrated value of FreshRoute lies in the way modules update one another. A forecast influences the timing and volume of listings, matched orders determine route planning, delivery events trigger payment release, and complaint outcomes feed back into reputation and future matching. Over time, the platform accumulates a structured dataset that links price expectations, operational execution, and verified quality outcomes. This closed-loop data model is essential for learning which interventions reduce waste and improve farmer returns.

The orchestration layer exposes APIs to the farmer app, buyer portal, and alert services. When connectivity is poor, mobile clients cache draft listings, local forecasts, and pending evidence uploads, then synchronize once a network becomes available. The same layer normalizes timestamps, validates order identifiers, and ensures that a complaint can only be filed against a delivered and confirmed order. These workflow rules are important because a technically strong

model is of limited value if surrounding operations are ambiguous or inconsistent.

To keep modules loosely coupled, FreshRoute uses shared identifiers and event-driven APIs. A produce lot is created at listing time and then propagates through forecast context, contract record, dispatch order, transport event, image evidence, and complaint case. Only high-value state transitions are committed to the permissioned ledger, while larger artifacts such as raw images, model features, and notification logs remain off-chain in application databases and object storage. This hybrid arrangement preserves auditability without forcing high-volume data onto the blockchain.

IV. EVALUATION AND RESULTS

FreshRoute consolidates multiple proposal-stage modules into deployable architecture; therefore, evaluation is conducted at both component level and system level. The assessment combines predefined evaluation metrics with preliminary experimental observations obtained from module-level testing and pilot-oriented simulations.

TABLE II. MODULE-LEVEL EVALUATION MEASURES

Module	Key Measures
AI Intelligence Forecast	WAPE, RMSE, MAE, Forecast accuracy, Prediction Time
Blockchain Marketplace	Match Acceptance Rate, Stock Utilization Rate, Time-to-Settlement, Escrow Default Rate
Optimized Logistics	Route Time, Delivery delay, Temperature Violation Flag, Freshness Cost.
Smart Quality & Compliance	Precision, Recall, F1- Score, Detection Accuracy, Quality Agreement rate.

A. Evaluation Design

The system is evaluated through a two-stage approach. First, component-level experiments are conducted for each module. Second, an integrated evaluation is performed through a controlled pilot involving selected farmer clusters, buyers, and transporters. A quasi-experimental setup is considered, where one group follows existing manual coordination practices while another utilizes FreshRoute-supported workflows. This enables comparative analysis of operational efficiency, reliability, and user experience.

B. Forecasting Performance

Forecasting accuracy is assessed using weighted absolute percentage error (WAPE), root mean square error (RMSE), and mean absolute error (MAE), with comparisons against baseline models such as Prophet and SARIMA [5] – [7].

Preliminary results indicate that the LSTM-based forecasting model achieves improved predictive performance, demonstrating lower error values compared to baseline approaches across selected fruit categories. The model maintains stability under seasonal variations through rolling time-window validation.

In addition to numerical accuracy, advisory usefulness is evaluated by measuring how frequently forecast outputs influence farmer decisions, including harvest timing, listing timing, and quantity offered. Observations suggest that forecast-driven advisories contribute to more informed decision-making and reduce uncertainty.

C. Marketplace and Matching Evaluation

Marketplace performance evaluation extends beyond ledger throughput to assess algorithmic efficacy and financial reliability. For the 5-factor matching algorithm, the

primary metric is the Match Acceptance Rate, which serves as empirical validation for the baseline heuristic weightings (e.g., the 0.35 spatial distance coefficient). To evaluate the efficacy of the "Smart Stock Split" concurrency model, the system tracks the Stock Utilization Rate, quantifying the reduction in orphaned inventory and measuring market liquidity improvements for small-volume buyers compared to traditional binary matching.

Financial disintermediation is evaluated by comparing traditional supply chain payment delays against the platform's Time-to-Settlement the latency between farm pickup and the smart contract's automated release of dynamically priced funds. Furthermore, the reliability of the trustless escrow is measured by the Escrow Default Rate and the activation frequency of the 48-hour stateful timeout. These metrics collectively validate the platform's ability to programmatically resolve buyer unresponsiveness, ensure immediate farmer liquidity, and eliminate counterparty risk without manual arbitration.

D. Logistics Optimization Performance

The logistics module is evaluated using route completion time, delivery delay, temperature violation frequency, and estimated freshness cost.

Results demonstrate that perishability-aware routing reduces delivery delays and improves vehicle utilization compared to non-optimized approaches. The integration of temperature and compatibility constraints contributes to minimizing spoilage risks during transportation. Lightweight telemetry signals, such as GPS and timestamp tracking, enable timely detection of route deviations and environmental violations.

E. Quality Verification and Complaint Validation

The quality module is evaluated using precision, recall, F1-score, detection accuracy, and agreement with human assessors. Preliminary results indicate that the computer vision model achieves high reliability in detecting fruit defects and differentiating quality levels. Complaint validation shows strong alignment between automated classification and human evaluation. Additionally, perceptual hashing effectively identifies duplicate or reused images, supporting fraud detection and improving trust in the complaint process.

F. Integrated System Evaluation

An integrated pilot is designed using fruit types with varying perishability profiles, such as banana, mango, and pineapple, to evaluate cross-module coherence under real-world operating conditions. The integrated evaluation demonstrates a noticeable reduction in transaction delays, along with improved coordination among stakeholders including farmers, buyers, and transport providers. In addition, the availability of verifiable evidence through the platform contributes to a decrease in the frequency of disputes, while simultaneously enhancing overall transparency and trust across the supply chain. Furthermore, forecast-guided selling shows potential in minimizing last-minute distress sales by enabling better planning decisions, and validated complaint outcomes contribute to improved matching behavior in subsequent transactions.

V. DISCUSSION

FreshRoute is designed for a context where infrastructure gaps matter as much as algorithmic choices. Three deployment decisions are therefore critical. First, the platform must degrade gracefully: forecasts and alerts should remain accessible under intermittent connectivity, and

logistics monitoring should work with sparse telemetry. Second, sensitive business data should not be published indiscriminately; the ledger should store contract events, hashes, and permissions, while bulk images and model features remain off chain. Third, fully automated enforcement should be avoided. Reputation penalties and suspensions are useful, but borderline cases still require human oversight, especially when losses may arise from handling faults beyond the farmer's control.

FreshRoute is best piloted in commodity clusters where supply is frequent and buyer demand is recurrent. Bananas, mango, and pineapple are suitable because they differ in perishability, handling sensitivity, and market rhythm. A useful pilot geography combines a production area, an aggregation point, and an urban demand node so that forecasting, matching, transport, and complaint handling are exercised end to end. Participant onboarding should begin with farmer registration, buyer verification, and minimal digital training. The platform turns complex backend processes into ordinary tasks such as creating listings, accepting orders, dispatching loads, confirming delivery, and uploading evidence, which is essential for building trust in smallholder environments.

A phased rollout is the most realistic path to adoption. The first phase introduces forecast advisories and digital listings without automated sanctions or complex routing. The second phase activates payment release logic and basic vehicle assignment once trading behavior stabilizes. The third phase adds telemetry-triggered rerouting and AI-backed complaint validation on active routes. This sequencing lowers adoption risk, reveals which module produces the greatest early value, and allows models to be recalibrated against observed user behavior. Pilot governance also matters: delivery failures may arise from producer faults, transport mishandling, or buyer-side storage conditions, so complaint outcomes should remain reviewable by local agricultural officers, cooperative leaders, or designated neutral assessors during early deployment.

Scaling beyond a pilot requires institutional anchoring. A cluster-based rollout through farmer organizations, collection centers, or regional agribusiness partnerships reduces training costs, creates enough transactional density for route optimization, and helps low-volume farmers who would otherwise struggle to benefit from direct trading. Interoperability is equally important: the system should expose simple APIs for price data, allow CSV-based onboarding where full integration is not yet possible, and maintain SMS and lightweight mobile workflows for participants without stable smartphone access.

FreshRoute can complement ongoing public efforts around packaging improvement, post-harvest management, and digital agriculture rather than compete with them. Data from crate-distribution programs, extension visits, or regional market information services could enrich both matching and advisory logic. In the longer term, platform output may also help policymakers identify where transport bottlenecks, recurrent complaint patterns, or weak cold-chain coverage create the highest waste, making the system valuable not only as a marketplace tool but also as a decision support layer for modernization.

The economic sustainability of the platform depends on keeping coordination costs below the value currently lost through spoilage and opaque intermediation. A feasible approach is to subsidize early adoption through institutional pilots and recover operational costs through modest transaction fees or logistics coordination charges, kept proportional and visibly linked to service value. Data

ownership should be handled carefully: personally identifiable information must be separated from analytical datasets, with clear visibility rules for contracts, images, and reputation events. If platform rules, ranking logic, or dispute processes become opaque, the technology risks reproducing the same asymmetries previously managed by intermediaries. Matching criteria, fee schedules, and complaint pathways should remain understandable to users and open to periodic auditing by participating institutions.

VI. LIMITATIONS AND FUTURE WORK

The first limitation concerns data availability and representativeness. Forecasting accuracy will vary by fruit, district, and season, and early pilots are likely to begin with sparse historical records. Missing observations, inconsistent transaction logs, and the absence of standardized post-harvest quality measurements may introduce bias into both predictive models and downstream evaluation. For this reason, future work should prioritize simple but disciplined data protocols before pursuing heavier model complexity.

A second limitation is causal attribution. Even if payment delays, spoilage incidents, or complaint frequency decline after deployment, it will not automatically follow that every improvement was produced by the platform. Seasonal supply changes, buyer behavior, infrastructure conditions, and training effects can all influence outcomes. Stronger future studies should therefore use baseline periods, comparison groups, and mixed quantitative and qualitative evidence so that operational change is not mistaken for causal impact.

A third limitation is organizational fairness. Reputation systems can improve accountability, but they can also penalize new entrants or actors who suffer losses outside their control. Future versions of FreshRoute should investigate weighted reputation updates, role-specific accountability, and review policies that distinguish between producer, transporter, and buyer-side failures. Complaint workflows should also be tested for strategic misuse so that the platform does not reproduce the same asymmetries it seeks to correct.

Future research should extend beyond technical performance to full economic and environmental assessment. This includes estimating changes in farmer revenue share, transaction costs, complaint resolution time, and avoidable spoilage under live operating conditions. It also includes quantifying whether better route selection and reduced distress selling creates measurable carbon and waste benefits. Such evidence will be essential if FreshRoute is to move from a promising integrated prototype to a scalable intervention for Sri Lanka's fruit sector.

VII. CONCLUSION

FreshRoute addresses a central weakness of Sri Lanka's fruit economy: post-harvest losses and value erosion emerge from disconnected decisions rather than from a single technical bottleneck. The paper has presented an integrated system architecture that brings together predictive demand forecasting, permissioned direct trade, freshness-aware logistics, and AI-based quality assurance. By linking these modules through shared identifiers, auditable workflows, and farmer-oriented interfaces, the platform converts isolated digital tools into a coherent operational system.

The immediate contribution is a grounded design and evaluation framework for future pilots. The broader contribution is conceptual: reducing waste and eliminating unnecessary middlemen requires simultaneous advances in prediction, trust, execution, and verification. Future work should therefore prioritize pilot deployment, user-centered

refinement, and empirical estimation of economic and environmental impact under real Sri Lankan operating conditions.

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IX. REFERENCES

- [1] L. Rajapaksha, D. M. C. C. Gunathilake, S. M. Pathirana, and T. N. Fernando, "Reducing post-harvest losses in fruits and vegetables for ensuring food security-Case of Sri Lanka," *MOJ Food Processing & Technology*, vol. 9, no. 1, pp. 7-16, 2021, doi: 10.15406/mojfpt.2021.09.00255.
- [2] S. Kamalakkannan et al., "Life Cycle Assessment of Food Loss Impacts: Case of Banana Postharvest Losses in Sri Lanka," *Procedia CIRP*, vol. 105, pp. 859-864, 2022, doi: 10.1016/j.procir.2022.02.142.
- [3] Ministry of Agriculture, Sri Lanka, "It was revealed that 19% vegetables and 21% fruits waste annually during transportation in Sri Lanka," Mar. 2023. [Online]. Available: <https://www.agrimin.gov.lk/web/index.php/en/en/news-scroll/2014-27-03-2023-3e>
- [4] National Institute of Post Harvest Management, "Introduction of safe packaging and transportation methods to reduce postharvest loss in perishable supply chain in Sri Lanka," Jan. 2021. [Online]. Available: <https://www.niphm.lk/activities/projects/p0009>
- [5] S. J. Taylor and B. Letham, "Forecasting at Scale," *The American Statistician*, vol. 72, no. 1, pp. 37-45, 2018, doi: 10.1080/00031305.2017.1380080.
- [6] T.-W. Yoo and I.-S. Oh, "Time Series Forecasting of Agricultural Products' Sales Volumes Based on Seasonal Long Short-Term Memory," *Applied Sciences*, vol. 10, no. 22, Art. no. 8169, 2020, doi: 10.3390/app10228169.
- [7] S. Siarni-Namini, N. Tavakoli, and A. Siarni Namin, "A Comparison of ARIMA and LSTM in Forecasting Time Series," in *Proc. 17th IEEE Int. Conf. on Machine Learning and Applications (ICMLA)*, 2018, pp. 1394-1401, doi: 10.1109/ICMLA.2018.00227.
- [8] A. Kamilaris, A. Fonts, and F. X. Prenafeta-Boldu, "The rise of blockchain technology in agriculture and food supply chains," *Trends in Food Science & Technology*, vol. 91, pp. 640-652, 2019, doi: 10.1016/j.tifs.2019.07.034.
- [9] E. Androulaki et al., "Hyperledger Fabric: A Distributed Operating System for Permissioned Blockchains," in *Proc. 13th EuroSys Conf.*, 2018, Art. no. 30, pp. 1-15, doi: 10.1145/3190508.3190538.
- [10] M. Cordeiro and J. C. Ferreira, "Beyond Traceability: Decentralised Identity and Digital Twins for Verifiable Product Identity in Agri-Food Supply Chains," *Applied Sciences*, vol. 15, no. 11, Art. no. 6062, 2025, doi: 10.3390/app15116062.
- [11] D. M. Utama, S. K. Dewi, A. Wahid, and I. Santoso, "The vehicle routing problem for perishable goods: A systematic review," *Cogent Engineering*, vol. 7, no. 1, Art. no. 1816148, 2020, doi: 10.1080/23311916.2020.1816148.
- [12] J. Naranjo-Torres et al., "A Review of Convolutional Neural Network Applied to Fruit Image Processing," *Applied Sciences*, vol. 10, no. 10, Art. no. 3443, 2020, doi: 10.3390/app10103443.
- [13] X. Gao et al., "Application of Advanced Deep Learning Models for Efficient Apple Defect Detection and Quality Grading in Agricultural Production," *Agriculture*, vol. 14, no. 7, Art. no. 1098, 2024, doi: 10.3390/agriculture14071098.
- [14] C. Zauner, "Implementation and Benchmarking of Perceptual Image Hash Functions," M.S. thesis, Upper Austria University of Applied Sciences, Hagenberg, Austria, 2010.